

Three-dimensional finite element simulation of magnetotelluric fields using unstructured grids

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1 Summary

The interpretation of an increasing number of three-dimensional data sets requires the simulation of the electromagnetic fields in three directions in space. Basing on Maxwell's equations different boundary value problems can be formulated in terms of electromagnetic potentials or fields including homogeneous or inhomogeneous boundary conditions.

The formulation of the equation of induction using vector and scalar potentials reduces the number of unknowns to four instead of six field components. Applying a secondary potential approach allows for the implementation of simple homogeneous Dirichlet boundary conditions. The boundary value problem is solved by means of the finite element method. We use quadratic Nédélec elements on unstructured tetrahedral grids that are well suited for incorporating arbitrary model geometries including surface and seafloor topography.

To expand the classic magnetotelluric frequency range towards lower periods ($T < 10^{-4}$ s) used by the Radio and Audio MT method for studying shallow conductivity structures displacement currents may need consideration. Beside the electric conductivity and permittivity, the presented finite element approach incorporates the magnetic permeability as model parameter that can be advantageous e.g. in the case of ore exploration and for studies of the earth's crust where basaltic rocks occur.

Keywords: Magnetotellurics, finite element method, numerical simulation

2 Introduction

The propagation of electromagnetic fields is governed by Maxwell's equations. They can be combined to yield the equation of induction in terms of the vector potential $\mathbf{A}$. The simulation of the secondary potential $\mathbf{A}_s$ minimizes the computational effort due to its local occurrence and the validity of homogeneous Dirichlet boundary conditions.

To solve the boundary value problem, we apply a finite element (FE) approach that allows for the efficient parametrization of arbitrary model geometries on unstructured tetrahedral grids. So-called vector or Nédélec elements are well suited to approximate the vector field $\mathbf{A}_s$ whose tangential components are continuous at element interfaces that represent possible jumps in the model parameters. Within one element, the vector potential $\mathbf{A}_s$ is described by polynomials.

The exploration of shallow conductivity structures requires the expansion of the classic magnetotelluric (MT) frequency range to shorter periods. Depending on the conductivity distribution the quasistatic approximation of Maxwell's equations might not be valid for periods shorter than 10^{-4} s. Hence, displacement currents need consideration. Furthermore, the presented FE algorithm incorporates the relative magnetic permeability μ_r as model parameter. By means of model studies we show the influence of displacement currents and the relative magnetic permeability μ_r on the apparent resistivity and the phase computed analytically for a homogeneous halfspace.

3 Equation of Induction

3.1 Maxwell's Equations

Assuming the harmonic time dependency $e^{i\omega t}$, the behaviour of the electric and the magnetic fields $\mathbf{E}$ and $\mathbf{H}$ is governed by Maxwell's equations of the form

$$\nabla \times \mathbf{H} = \mathbf{j} + i\omega \mathbf{D}, \quad (1)$$

$$\nabla \times \mathbf{E} = -i\omega \mathbf{B}, \quad (2)$$

$$\nabla \cdot \mathbf{D} = \rho,$$

$$\nabla \cdot \mathbf{B} = 0.$$

The eddy current density $\mathbf{j}$, the displacement current density $\mathbf{D}$, and the magnetic flux density $\mathbf{B}$ are combined with the electromagnetic fields by Ohm's law and the constitutive relations, respectively,

$$\mathbf{j} = \sigma \mathbf{E}, \quad \mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E}, \quad \text{and} \quad \mathbf{B} = \mu_0 \mu_r \mathbf{H}, \quad (3)$$

with the electric conductivity σ , the electric field constant $\epsilon_0 = 8.854 \cdot 10^{-12} \text{As/Vm}$, the relative electric permittivity ϵ_r , the magnetic field constant $\mu_0 = 4\pi \cdot 10^{-7} \text{Vs/Am}$, and the relative magnetic permeability μ_r .

3.2 Electromagnetic Potentials

The divergence-free field $\mathbf{B}$ can be expressed as curl of the vector potential $\mathbf{A}$

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (4)$$

Since

$$\nabla \times (\mathbf{E} + i\omega \mathbf{A}) = 0, \quad (5)$$

we can introduce the scalar potential V so that

$$\mathbf{E} = -\nabla V - i\omega \mathbf{A}. \quad (6)$$

Applying $\nabla \times$ on eq. (1), inserting Ohm's law and the constitutive relations (eqs. 3) yields

$$\nabla \times \mu^{-1} \nabla \times \mathbf{A} + (i\omega\sigma - \omega^2\epsilon)\mathbf{A} + (\sigma + i\omega\epsilon)\nabla V = 0. \quad (7)$$

Choosing

$$\tilde{\mathbf{A}} = \mathbf{A} - \nabla \Psi \quad \text{and} \quad \tilde{V} = V - \dot{\Psi}$$

with the gauge condition $\Psi = -iV/\omega$, we obtain

$$\tilde{\mathbf{A}} = \mathbf{A} - \frac{i}{\omega} \nabla V \quad \text{and} \quad \tilde{V} = 0 \quad (8)$$

that determine the same electromagnetic fields as $\mathbf{A}$ and V (cf. eqs 4 and 6). Using eq. (8), eq. (7) can be rearranged into an elliptic second-order partial differential equation for $\tilde{\mathbf{A}}$

$$\nabla \times \mu^{-1} \nabla \times \tilde{\mathbf{A}} + (i\omega\sigma - \omega^2\epsilon)\tilde{\mathbf{A}} = 0.$$

3.3 Secondary Potential Approach

The separation of the potential $\tilde{\mathbf{A}}$ (in the following: $\mathbf{A}$) into a normal ($\mathbf{A}_n$) and an anomalous ($\mathbf{A}_s$) contribution $\mathbf{A} = \mathbf{A}_n + \mathbf{A}_s$ results in a differential equation for $\mathbf{A}_s$

$$\nabla \times \mu^{-1}(\nabla \times \mathbf{A}_s - \mu_s \mathbf{H}_n) + (i\omega\sigma - \omega^2\epsilon)\mathbf{A}_s = (\sigma_s + i\omega\epsilon_s)\mathbf{E}_n \quad (9)$$

with

$$\epsilon = \epsilon_n + \epsilon_s, \quad \sigma = \sigma_n + \sigma_s, \quad \mu = \mu_n + \mu_s$$

and

$$\nabla \times \mathbf{E}_n = -i\omega\mu_n \mathbf{H}_n, \quad \nabla \times \mathbf{H}_n = (\sigma_n + i\omega\epsilon_n)\mathbf{E}_n.$$

The normal electromagnetic fields $\mathbf{E}_n$ and $\mathbf{H}_n$ are computed for a one-dimensional layered halfspace with parameter distributions σ_n , μ_n , and ϵ_n by Wait's algorithm (Wait, 1953).

3.4 Boundary Value Problem

Considering eq. (9) in the domain Ω with the outer boundary Γ_D and all internal boundaries Γ_{int} for which the conditions of continuity for the magnetic field are valid yields the boundary value problem: Find $\mathbf{A}_s$, so that

$$\nabla \times \mu^{-1}(\nabla \times \mathbf{A}_s - \mu_s \mathbf{H}_n) + (i\omega\sigma - \omega^2\epsilon)\mathbf{A}_s = (\sigma_s + i\omega\epsilon_s)\mathbf{E}_n \quad \text{in } \Omega, \quad (10)$$

$$\mathbf{A}_s = 0 \quad \text{on } \Gamma_D, \quad (11)$$

$$\mathbf{n}_1 \times \mathbf{H}_1 - \mathbf{n}_2 \times \mathbf{H}_2 = 0 \quad \text{on } \Gamma_{int} \quad (12)$$

with the outward unit normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$.

4 Finite Element Method

4.1 Weak Form

An equivalent formulation of eq. (10) as an inner product $(\mathbf{v}, \mathbf{u}) = \int_{\Omega} \bar{\mathbf{v}} \cdot \mathbf{u} dV$ with a complex vector-valued test function $\mathbf{v}$ from the function space V yields the so-called weak form of the boundary value problem: Find $\mathbf{A}_s \in U$, so that

$$\begin{aligned} \int_{\Omega} (\mu^{-1}(\nabla \times \mathbf{A}_s - \mu_s \mathbf{H}_n) \cdot \nabla \times \bar{\mathbf{v}} + (i\omega\sigma - \omega^2\epsilon)\mathbf{A}_s \cdot \bar{\mathbf{v}}) dV + \underbrace{\int_{\partial\Omega} \mathbf{n} \times (\mu^{-1}(\nabla \times \mathbf{A}_s - \mu_s \mathbf{H}_n) \cdot \bar{\mathbf{v}}) dS}_0 \\ = \int_{\Omega} (\sigma_s + i\omega\epsilon_s)\mathbf{E}_n \cdot \bar{\mathbf{v}} dV \quad \forall \mathbf{v} \in V, \end{aligned} \quad (13)$$

with

$$U := \{\mathbf{A}_s \in H(\text{curl}, \Omega) : \mathbf{A}_s \equiv 0 \text{ on } \Gamma_D\},$$

$$V := \{\mathbf{v} \in H(\text{curl}, \Omega) : \mathbf{v} \equiv 0 \text{ on } \Gamma_D\} \quad (14)$$

and

$$H(\text{curl}, \Omega) := \{\mathbf{u} \in (L^2(\Omega))^3, \nabla \times \mathbf{u} \in (L^2(\Omega))^3\}.$$

Applying the boundary conditions in eqs. (11) and (12), the boundary integral in eq. (13) vanishes.

4.2 Finite Element Analysis

A discrete approximation $\mathbf{A}_s^h \in U_h$ of $\mathbf{A}_s \in U$ arises from the linear combination of N real vector-valued basis functions $\phi_i \in U_h$ ($i = 1, \dots, N$) with the complex coefficients a_i ($i = 1, \dots, N$):

$$\mathbf{A}_s^h = \sum_{i=1}^N a_i \phi_i. \quad (15)$$

Using the discrete test functions $\mathbf{v}_i = \phi_i$ the boundary value problem can be written as matrix-vector-equation

$$\tilde{\mathbf{K}} \mathbf{A}_s = \mathbf{L} \quad (16)$$

whereas

$$\begin{aligned} K_{i,j} &= \int_{\Omega} (\mu^{-1} (\nabla \times \phi_i - \mu_s \mathbf{H}_n) \cdot \nabla \times \bar{\phi}_j + (i\omega\sigma - \omega^2\epsilon) \phi_i \cdot \bar{\phi}_j) dV, \\ L_i &= \int_{\Omega} (\sigma_s + i\omega\epsilon_s) \mathbf{E}_n \cdot \bar{\phi}_i dV. \end{aligned} \quad (17)$$

The domain Ω is decomposed into tetrahedra. On each tetrahedron piecewise quadratic basis functions ϕ_i are assumed. Their degrees of freedom are associated with the edges and the faces of the tetrahedral finite element (Fig. 1). This type of finite elements, the so-called vector or Nédélec elements, is especially suitable for the discretization of vector fields that show continuity in their tangential components.

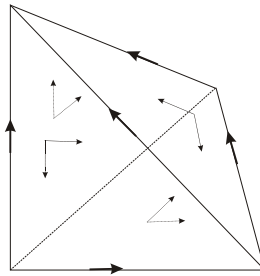


Fig. 1: Graphical representation of the degrees of freedom for quadratic Nédélec elements on a tetrahedron.

5 Influence of Permittivity and Permeability

5.1 Displacement Currents

Usually, in the simulation of the MT method displacement currents are neglected due to the classic frequency range not exceeding 1 kHz. To study shallow conductivity structures, the Radio and Audio MT method applies frequencies up to several MHz. Fig. 2 presents the apparent resistivities and phases for a homogeneous halfspace of $1000 \Omega \text{ m}$ (left) and $10000 \Omega \text{ m}$ (right), respectively, computed including ($\epsilon_r = 1$, 'o') and without (quasistatic, '+') displacement currents. Significant deviations are obvious for frequencies higher than 10 kHz especially for the more resistive halfspace.

5.2 Magnetic Rock Properties

So far, the magnetic rock properties have not been incorporated in electromagnetic applications. However, in addition to the electric conductivity σ , the relative magnetic permeability μ_r might yield geological information especially in the field of ore exploration and studies of regions where basaltic rocks occur. Fig. 3 displays the dependency of the apparent resistivity and phase on μ_r . $\mu_r > 1$ results in a higher induced current density (cf. eqs.

(2) and (3)) and therefore yields a lower apparent resistivity. The phase features no anomalous behaviour. Note, that there is no frequency dependency at all.

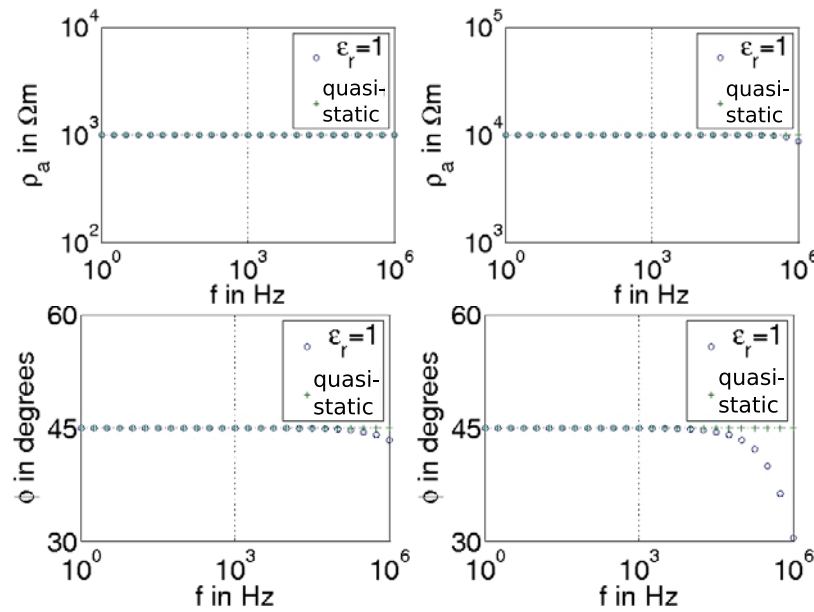


Fig. 2: Apparent resistivities and phases for a homogeneous halfspace of $1000 \Omega m$ (left) and $10000 \Omega m$ (right).

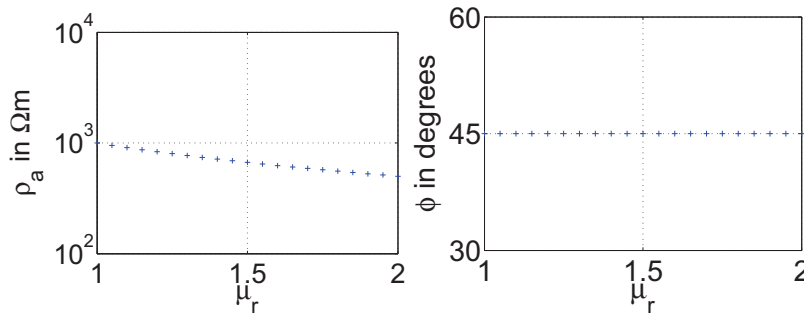


Fig. 3: Apparent resistivities and phases over μ_r varying in the range of $1 \leq \mu_r \leq 2$ for a homogeneous halfspace of $\rho = 1000 \Omega m$.

6 Comparison with an FD approach

We present the simulation results for the 3D-2 COMMEMI model (Weaver and Zhdanov, 1997) shown in Fig. 4. The apparent resistivities ρ_{xy} and ρ_{yx} as well as the phases ϕ_{xy} and ϕ_{yx} on the earth's surface at $x = 0$ computed by our FE algorithm and Mackie's finite difference (FD) code (Mackie, Madden and Wannamaker, 1993) are displayed in Fig. 5.

The anomalous resistivities of the two bodies are obvious. The steep slopes in ρ_{yx} and ϕ_{yx} (fig. 5, right) are due to the discontinuity of the normal component of the electric field whereas the tangential component used to determine ρ_{xy} and ϕ_{xy} (fig. 5, left) is continuous. The results for both of the approaches are in good agreement. However, we expect our FE code to yield the more accurate solution concerning the advantages of the FE method approximating electromagnetic fields especially at high parameter contrasts that we have experienced for two-dimensional simulations.

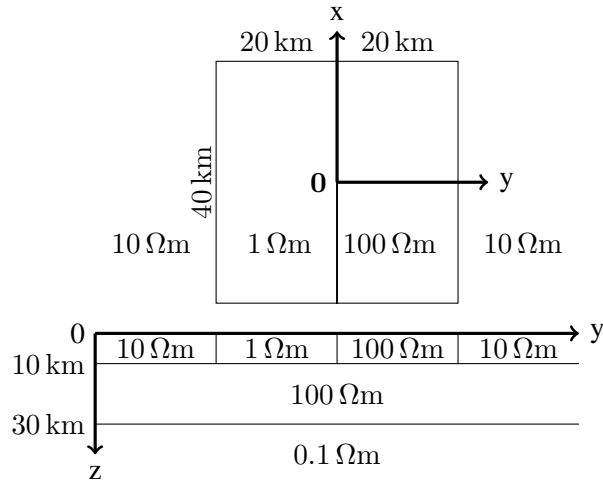


Fig. 4: COMMEMI model 3D-2.

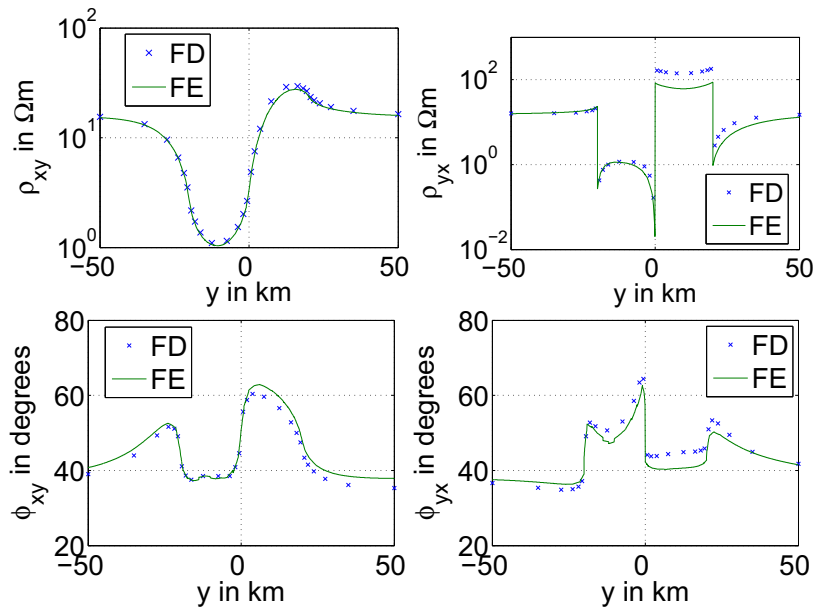


Fig. 5: Apparent resistivities and phases for the COMMEMI model 3D-2 at a period of $T = 100$ s.

7 Conclusions

The presented secondary potential approach has proven to be well suited for the three-dimensional simulation of electromagnetic fields. The local occurrence of the anomalous vector potential allows for the implementation of homogeneous Dirichlet boundary conditions for relatively small models hence minimizing the computational effort in terms of memory. However, future efficiency studies will include formulations of the boundary value problem for the total potential and the electromagnetic fields.

The approximation of the magnetic vector potential by quadratic Nédélec elements on unstructured tetrahedral grids achieves a satisfying accuracy in comparison with other numerical approaches. We intend to improve the efficiency and accuracy of our algorithm by employing an adaptive mesh refinement in connection with an error estimator function.

Model studies clearly show that the consideration of displacement currents is essential for frequencies higher than 10 kHz. Additionally, magnetic rock properties, namely $\mu_r > 1$, can influence the apparent resistivity and phase significantly. This might be of interest with regard to the hypothesis of the second-order magnetic phase transition taking place at medium depths of the earth's crust (Kiss, Szarka and Prácser, 2005).

References

- Kiss, J., Szarka, L. and Prácser, E. (2005). Second-order magnetic phase transition in the earth. *Geophys. Res. Lett.*, 32(L24310).
- Mackie, R. L., Madden, T. R. and Wannamaker, P. E. (1993). Three-dimensional magnetotelluric modeling using difference equations - theory and comparisons to integral equation solutions. *Geophys.*, 58, 215-226.
- Wait, J. R. (1953). Propagation of radio waves over a stratified ground. *Geophysics*, 20, 416-422.
- Weaver, J. and Zhdanov, M. (1997). Methods for modelling electromagnetic fields. *J. Appl. Geophys.*, 37, 133-271.